Beyond the Training Distribution: A Survey of Reasoning Generalization in Large Language Models

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Abstract

Large language models (LLMs) have achieved strong performance on mathematical, logical, and scientific reasoning tasks. However, this progress is commonly measured on benchmarks that remain close to the distributions encountered during training. Small changes in notation or problem structure, as well as deeper reasoning or new domains, can lead to substantial performance degradation, raising a central question: *when and how can LLMs generalize their reasoning beyond their training distributions?* This paper presents a comprehensive survey of reasoning generalization in LLMs. We organize existing work into four complementary pillars: (1) training methods that encourage transferable reasoning, (2) inference strategies that adapt computation at test time, (3) architectural designs that support reusable operations, and (4) analyses that characterize when and why generalization succeeds or fails. Building on this taxonomy, we compare the assumptions, evaluation settings, and generalization claims of representative approaches. We further summarize the main empirical findings and theoretical results, and discuss the evaluation practices and open challenges that shape this emerging research direction.

1 Introduction

Recent advances in prompting, post-training, and test-time computation have improved the reasoning capabilities of large language models (LLMs). Models can now solve difficult mathematical problems, construct multi-step proofs, and use external tools. These advances raise a central question: when and how can LLMs generalize their reasoning beyond their training distributions?

Strong benchmark performance does not resolve this question. Accuracy can vary across equivalent numerical versions of a problem and decline when irrelevant clauses are added (Mirzadeh et al., 2025). Models also struggle with deeper or wider proof trees (Opedal et al., 2025) and unfamiliar combinations of scientific domains (Gong et al., 2026). Similar failures occur under symbol remapping, longer inputs, new task compositions, and changed tool interfaces. These findings suggest that success on held-out examples alone does not establish that a model has learned a reusable reasoning procedure.

Assessing reasoning generalization requires a clear account of what is "unseen." A task absent from supervised fine-tuning may still resemble pre-training data, while extra samples or tool calls may extend system performance without improving the base model. Studies also differ in scope, ranging from evaluations of frontier LLMs to experiments on small transformers and formal results under restricted assumptions. Comparisons therefore need to state what the model has encountered, what changes at test time, and which components remain fixed.

Efforts to improve reasoning generalization mainly span training, inference, and architecture. Training methods seek transferable procedures through data, objectives, and curricula. Inference methods use structured reasoning, additional computation, verification, and external memory to guide reasoning at test time. Architectural studies investigate recurrent depth, information routing, and recurrent memory. Analytical studies examine the resulting behavior, internal mechanisms, and theoretical limits. Existing surveys cover broad NLP generalization (Hupkes et al., 2023), out-of-distribution learning (Liu et al., 2021), inductive reasoning (Chen et al., 2025b), or generalization in language-model agents (Zhang et al., 2025b). Appendix B.2 compares our scope with related surveys of latent reasoning, logical reasoning, and planning. Our focus is the transfer of reasoning procedures across explicit distribution shifts.

This survey brings these research directions to-

gether. As shown in Figure 1, we organize the literature into four pillars: *Training for Generalization* (§3), *Inference for Generalization* (§4), *Architecture for Generalization* (§5), and *Analysis of Generalization* (§6). Together, these pillars organize methods that support reasoning generalization and evidence about the conditions under which it succeeds or fails.

Our key contributions are threefold: (1) **Systematic Taxonomy.** We present a unified taxonomy that organizes studies of reasoning generalization into four pillars (Figure 1), clarifying their scope and objectives. (2) **In-depth Discussion.** Building on this taxonomy, we analyze representative works, comparing training strategies, inference procedures, architectural designs, and behavioral, mechanistic, and theoretical evidence to examine when and how reasoning generalizes. (3) **Challenges and Future Directions.** We identify key challenges in transfer, length and depth extrapolation, and interpretability, and outline promising research directions in transferable training, stable recurrent computation, and verification (§7).

2 Preliminaries

Definitions. Let x denote a reasoning problem, y its answer, and a the inference procedure applied by a model with parameters θ. The resulting system is written as $f_{\theta,a}(x)$, with expected risk $R_P(\theta, a) = \mathbb{E}_{(x,y)\sim P}[\ell(f_{\theta,a}(x), y)]$ on a distribution P. Evaluation is in-distribution (ID) when training and test examples follow the same relevant data-generating process, and out-of-distribution (OOD) when a specified property of that process changes (Liu et al., 2021). When the two losses are comparable, the transfer gap is $R_{\mathrm{OOD}}(\theta, a) - R_{\mathrm{ID}}(\theta, a)$. We use *reasoning generalization* to describe the ability to preserve a reasoning procedure under such a change. This includes compositional generalization to new combinations of familiar primitives (Lake and Baroni, 2018) and length generalization to longer or deeper instances (Anil et al., 2022).

Overview. Following Figure 1, *Training for Generalization* (§3) reviews how parameter learning supports transfer, *Inference for Generalization* (§4) covers test-time reasoning procedures, including verifier-guided updates to generator weights, and *Architecture for Generalization* (§5) studies designs that support reusable computation. Beyond these methods, *Analysis of Generalization* (§6) examines what transfers and why. Methods spanning several pillars are discussed under their primary source of generalization, and §7 outlines open challenges and future directions.

3 Training for Generalization

Training-based methods update model parameters to improve reasoning generalization under shifts in surface form, composition, length, difficulty, domain, or language. The central challenge is that training can reinforce any pattern that fits the training data, including shortcuts that break under shift (Dziri et al., 2023; Mirzadeh et al., 2025). We therefore organize these methods by training stage and, within each stage, by the component they modify. Appendix D provides the training-data inventory (Table 6), evaluation benchmarks (Table 10), and survey-wide evaluation metrics (Table 12); the training protocols and reported results are documented in Appendix D.3.

3.1 Pre-Training

At the earliest stage, pre-training provides the model with its initial knowledge and reasoning primitives (i.e., basic operations). By the component they modify, methods fall into data curation and optimizer design. Data curation controls what the corpus contains, covering the primitives that later compositions require (Zhang et al., 2025a) and raising the ratio of inferred to stated facts so that the model acquires rules rather than memorizing (Wang et al., 2024a; Abramov et al., 2025). Optimizer design instead changes how the model fits the same corpus, either by aligning gradients across data sources (Chen et al., 2026a) or by using small initialization and strong weight decay to favor primitives over memorization (Zhang et al., 2024d, 2025c), although generalizing solutions reached by training far beyond overfitting can transfer poorly to new knowledge (He et al., 2026).

3.2 Mid-Training

Subsequently, mid-training inserts a stage between pre-training and task-specific adaptation, which narrows the gap between the pre-training and post-training distributions and supplies skills required by the target tasks; without it, a model lacking such a skill tends to memorize the tasks during post-training rather than acquire it (Zhang et al., 2025a; Jin et al., 2025). By the objective of this stage, we distinguish two common approaches: (1)

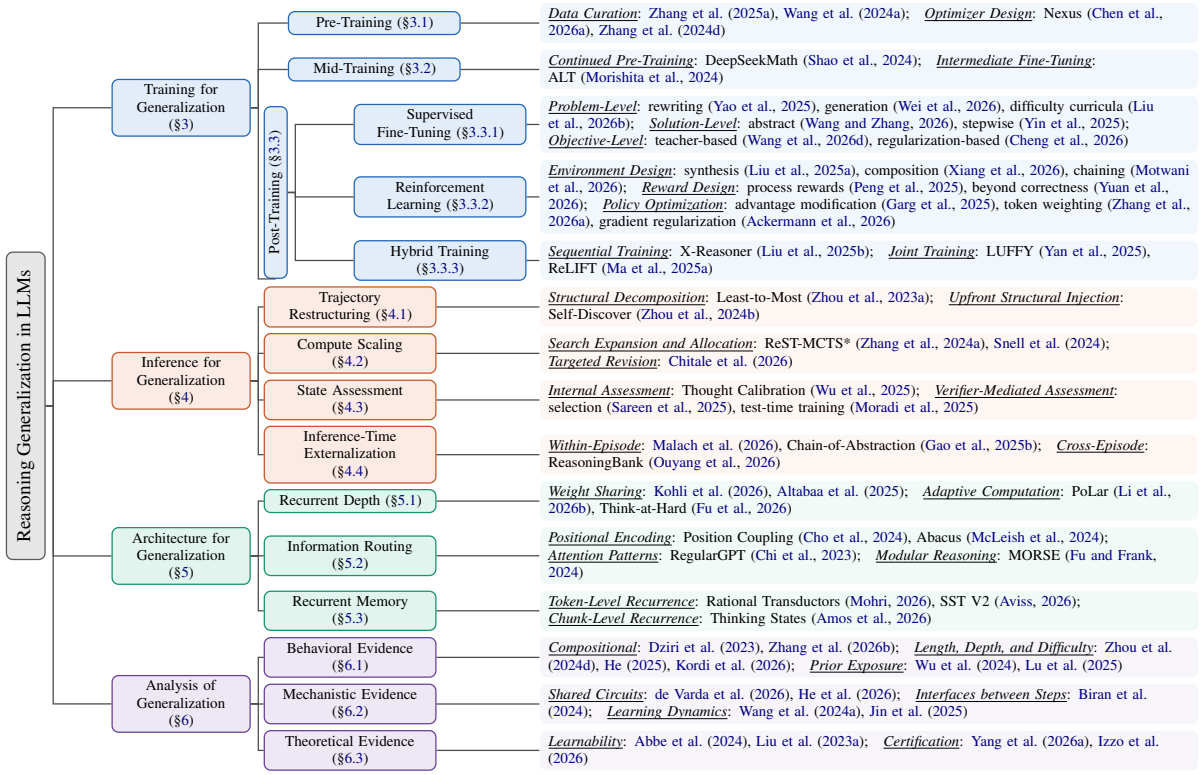


Figure 1: Taxonomy of reasoning generalization in LLMs, with representative works discussed in the main text.

Continued Pre-Training methods, which apply next-token prediction to raw text from a domain under-represented in pre-training, directly narrowing this gap (Shao et al., 2024; Fesser et al., 2026). (2) **Intermediate Fine-Tuning** methods, which train on solved problems that require no domain knowledge, such as program-generated deduction or symbolic problems, so that the acquired reasoning transfers to other domains and to realistic problems in natural language (Morishita et al., 2024; Cao et al., 2026). Beyond such direct transfer, this stage also prepares later training, after which the model generalizes better to OOD tasks and composes separately learned skills into novel combinations (Shrestha et al., 2025; RRV et al., 2026; Cheng et al., 2025b).

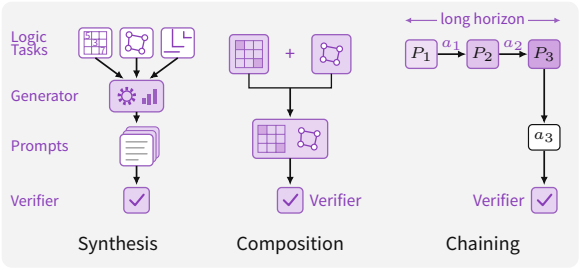
3.3 Post-Training

Post-training fine-tunes the model on reasoning tasks and is where the ID/OOD gap most easily opens: a model accurate on training tasks can still fail on rephrased, longer, or out-of-domain problems (Mirzadeh et al., 2025; Opedal et al., 2025; Gong et al., 2026). We cover Supervised Fine-Tuning (SFT), Reinforcement Learning (RL), and Hybrid Training.

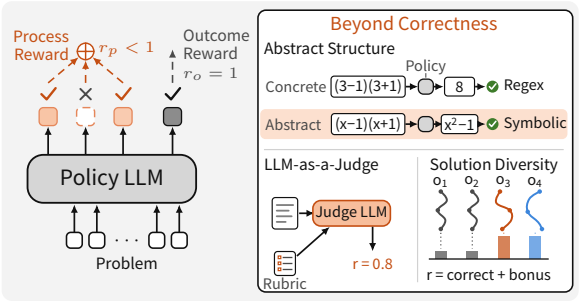
3.3.1 Supervised Fine-Tuning

SFT trains the model to imitate demonstrations, i.e., step-by-step solutions to training problems, and thus tends to memorize their spurious surface correlations, becoming brittle under shift (Chu et al., 2025; Peng et al., 2026). Accordingly, SFT methods intervene at the level of the problems, the solutions, or the objective.

Problem-Level Methods in SFT. To begin with, this line of work modifies the training problems so that the shift anticipated at test time is already represented in training, leaving no single surface pattern for the model to memorize. (i) **Problem Rewriting** presents each problem in variants that differ only in solution-irrelevant details, teaching the model that the answer stays fixed under surface shift (Yao et al., 2025; Gui and Ji, 2025). (ii) **Problem Generation** creates problems for regions the original data does not cover, such as rare skill combinations or slightly harder instances, which strengthens compositional and length generalization (Wei et al., 2026; Lee et al., 2025). (iii) **Difficulty Curriculum** orders or selects training problems by difficulty, generalizing to harder or OOD problems (Liu et al., 2026b,d); its benefit nevertheless depends on the shift, as difficulty ordering shows no robust advantage over random sampling



(a) Environment Design



(b) Reward Design

Figure 2: Environment and reward design in RL: (a) building verifiable training problems, and (b) process, outcome, and beyond-correctness rewards.

for compositional generalization (Mordig et al., 2026).

Solution-Level Methods in SFT. In contrast, these methods rewrite the solutions that the model imitates. The central insight is that imitation generalizes only as far as its target does, so only a target exposing a reusable procedure transfers to new problems (Peng et al., 2026; Matsutani et al., 2026). (i) **Abstract Solutions** remove problem-specific details and teach a general strategy or rule before applying it to a specific problem, addressing surface and domain shift (Wang and Zhang, 2026; Peng et al., 2026). (ii) **Stepwise Solutions** write out every step of a rule allowing it to run for more steps or combine with others and thus to cope with length and compositional shift (Hu et al., 2024; Wang et al., 2026a; Hu et al., 2025b; Yin et al., 2025; Hua et al., 2025).

Objective-Level Methods in SFT. Inspired by knowledge distillation (Hinton et al., 2015) and invariance-based regularization from domain generalization (Arjovsky et al., 2019), these methods modify the loss so that what would not hold under shift contributes less to the update. (i) **Teacher-based Objectives** match the output distribution of a teacher on the student's own solutions instead of imitating a single written solution, which removes the mismatch with the expert solutions seen in training and enables transfer across domains, reasoning horizons, and languages (Li et al., 2026a; Wang et al., 2026d). More selective variants restrict this matching to tokens or spans the student can support, which yields further OOD gains (Jukić and Titov, 2026; Zhao et al., 2026c). (ii) **Regularization-based Objectives** constrain or reweight the imitation loss, keeping only gradient directions shared by logically equivalent problems or reweighting samples, including wrong-answer ones, and outperform standard SFT on OOD problems (Cheng et al., 2026; Tian et al., 2026). Despite these gains, a teacher given rich privileged information can suppress the student's uncertainty and degrade accuracy on unseen problems (Kim et al., 2026).

3.3.2 Reinforcement Learning

Unlike SFT, RL replaces reference solutions with a reward, usually whether a checker accepts the final answer, and updates the model toward higher-reward samples (Shao et al., 2024). Without imitation, RL reportedly generalizes to problem variants and domains where SFT degrades (Chu et al., 2025; Huan et al., 2025). Yet RL generalization is bounded: OOD gains seem to appear only when the target task aligns with pre-training (Ni et al., 2025), and accuracy collapses at a few times the training depth (Wang et al., 2026c). We further group RL methods by the component they modify: the environment, the reward, or the policy update (Figures 2 and 3).

Environment Design. First, these methods change the problems and checkers that the model practices on (Figure 2a) and hence control which tasks, difficulties, and reasoning horizons training covers. (i) **Synthesis** employs generators that turn logic tasks into checkable problems of controlled difficulty, with gains that extend to OOD puzzles and other domains (Liu et al., 2025a; Chen et al., 2025a); along this line, even environments built on fictional knowledge or games transfer to real-world problems (Kabra et al., 2026; Feng et al., 2026). (ii) **Composition** merges existing environments into tasks that require several skills at once, generalizing to unseen benchmarks (Xiang et al., 2026). (iii) **Chaining** feeds the answer of each short problem into the next, letting a single verifiable answer span a longer horizon and carry over to longer and harder problems (Motwani et al., 2026).

Reward Design. Second, instead of changing the problems, another family changes what is rewarded (Figure 2b). The central insight is that an outcome-only reward credits any path to a correct answer, including shortcuts that hold only in distribution; thus, this family also rewards properties of the reasoning itself. (i) **Process Rewards** score each intermediate step (r_p), not merely the final answer judged by outcome rewards (r_o), favoring a correct procedure over such shortcuts (Salimi et al., 2026; Peng et al., 2025). (ii) **Rewards beyond Correctness** score properties of a solution other than its answer, in three forms. *Abstract structure* rewards favor solutions that capture the problem symbolically regardless of its names and narrative, narrowing the OOD gap of SFT and increasing robustness to surface perturbations (Yuan et al., 2026; Gao et al., 2025a). An *LLM-as-a-judge* replaces or complements rule-based checkers, extending rewards to answers that rules cannot verify (Ma et al., 2025b; Bhattarai et al., 2026) and raising accuracy outside the training domain (Zha et al., 2025; Ma et al., 2025b; Bhattarai et al., 2026). Lastly, *solution diversity* rewards grant a bonus to solutions that differ from the rest of the group, yielding higher out-of-domain accuracy (Hu et al., 2026b; Zhou et al., 2025; Maniparambil et al., 2026).

Policy Optimization. Finally, rather than changing the problems or the reward, these methods modify the policy update itself, so that it does not reinforce shortcuts that hold only on the training data. Figure 3 illustrates three such modifications to the standard GRPO update (Shao et al., 2024). (i) **Advantage Modification** reshapes the advantage, i.e., how much a rollout outscores its group, ensuring that incorrect rollouts with partial credit never receive a positive update, strengthening OOD transfer (Garg et al., 2025). (ii) **Token Weighting** assigns each token its own credit according to how strongly it signals a correct rather than an incorrect rollout (Zhang et al., 2026a; Xu et al., 2026), avoiding the OOD degradation of standard updates (Xu et al., 2026). (iii) **Gradient Regularization** penalizes the gradient norm to steer training toward flatter regions where the reward is more accurate, which curbs reward hacking, i.e., exploiting flaws of the reward (Ackermann et al., 2026).

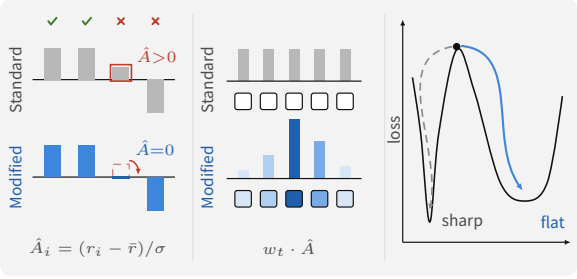


Figure 3: Policy optimization in RL: standard GRPO updates (gray) versus modified ones (blue).

3.3.3 Hybrid Training

Alternatively, hybrid methods combine demonstrations that introduce procedures the model cannot learn on its own with rewards that recover OOD accuracy reduced during SFT (Jin et al., 2025; Fesser et al., 2026). Depending on how the two signals are scheduled, we distinguish (i) **Sequential Training**, i.e., SFT followed by RL, which transfers reasoning to other domains, modalities, and languages (Liu et al., 2025b; Gurgurov et al., 2026); since excessive SFT leaves RL unable to improve, Rejuvenation (Liu et al., 2026c) partially resets the model toward its base weights before RL, restoring OOD gains. (ii) **Joint Training**, which combines the two signals within one stage, in the sampled solutions (Yan et al., 2025; Huang et al., 2025d), in the loss (Fu et al., 2025; Hu et al., 2026a), or by gating SFT and RL updates on rollout success (Lv et al., 2025; Ma et al., 2025a; Zhu et al., 2026), with several methods reporting higher OOD accuracy than RL alone (Yan et al., 2025; Fu et al., 2025; Lv et al., 2025). This advantage is challenged by Limozin et al. (2026): with corrected baselines, sequential training surpasses these joint methods, including OOD (Appendix D.3, Table 7).

4 Inference for Generalization

Autoregressive generation can compound errors on OOD tasks. Test-time methods target different parts of inference: structuring trajectories (Trajectory Restructuring), exploring alternatives (Compute Scaling), assessing reasoning reliability (State Assessment), and using external state or tools (Inference-Time Externalization). For fixed-weight methods, inference scaling cannot synthesize algorithmic reasoning entirely absent from training (Tong et al., 2026). Against this backdrop,

Table 11 summarizes the empirical evidence for these methods and their key caveats.

4.1 Trajectory Restructuring

These methods organize free-form generation without updating model weights. **Structural decomposition** breaks complex reasoning into manageable steps (Zhou et al., 2023a; Zhang et al., 2024c), using independently executed micro-agents (Meyerson et al., 2025) or predefined atomic skills (Chen et al., 2024). **Upfront structural injection** constrains generation in advance by restricting valid answers (Ma et al., 2026), constructing task-specific procedures (Zhou et al., 2024b), or synthesizing query-specific demonstrations (He et al., 2024b). The latter addresses out-of-demonstration generalization, which should not be equated with an explicit shift from the pretraining distribution.

4.2 Compute Scaling

When one trajectory is insufficient, extra test-time compute can explore alternatives or revise errors. **Search expansion and allocation** uses process-level signals to guide exploration: ReST-MCTS* (Zhang et al., 2024a) combines a Process Reward Model (PRM) with Monte Carlo Tree Search (MCTS) to score partial solutions, while MCTS can also improve policy and reward models for finer inference guidance (Chan et al., 2025). Adaptive methods allocate compute according to estimated problem difficulty (Snell et al., 2024), although this alone does not establish OOD generalization. **Targeted revision** concentrates computation on faulty reasoning steps rather than regenerating an entire solution (Chitale et al., 2026).

4.3 State Assessment

State assessment estimates the reliability of intermediate reasoning to guide selection or further computation. **Internal assessment** draws on residual-stream trajectories (Damirchi et al., 2026), uncertainty-progress alignment (Yan et al., 2026), or thought calibration for risk-controlled stopping (Wu et al., 2025). Online calibration can further update reliability estimates under distribution shift (Zhou et al., 2026). **Verifier-mediated assessment** uses auxiliary models for candidate selection or adaptation: joint reasoner–verifier training supports inference-time selection in evaluated mathematical and physics tasks (Sareen et al., 2025), while verifier-guided test-time training updates generator weights using high-confidence outputs on unlabeled target data (Moradi et al., 2025). However, PRMs can generalize poorly to harder problems or solutions from different models (Zhu et al., 2025a).

4.4 Inference-Time Externalization

Externalization gives inference access to working state or computation beyond the model's internal state. **Within-episode externalization** uses interactive memory to support length extrapolation (Malach et al., 2026), separates abstract reasoning from tool-supplied values (Gao et al., 2025b), and supports tool selection or restraint (Alazraki and Rei, 2025; Qian et al., 2025). These gains do not by themselves establish that the model generalizes without external resources. **Cross-episode externalization** stores strategies distilled from prior successes and failures in persistent memory (Ouyang et al., 2026); modular sub-instructions and learned selection mechanisms support memory retrieval (Tong and Gong, 2026), although cross-domain transfer is not directly established by the main evaluations.

5 Architecture for Generalization

Architectural approaches change the computation available to a model: how it reuses transformations, connects representations, or carries state forward. Such changes provide inductive biases that can support reasoning generalization by allowing a model to reuse learned operations when problems grow longer or combine familiar steps in new ways.

5.1 Recurrent Depth

Recurrent depth repeatedly applies a shared transformation, increasing computation without a separate parameter set at each depth (Geiping et al., 2025; Zeng et al., 2026; Zhu et al., 2025c). This can support length generalization when longer problems require more iterations of the same operation, as shown on controlled algorithmic tasks (Fan et al., 2025). Designs differ in which computation is reused and how its iteration count is adjusted across inputs or tokens. **Weight Sharing.** Shared recurrent blocks support generalization to deeper compositions of facts (Kohli et al., 2026). On computation graphs, recurrence combined with latent supervision, discretization, and

error correction enables generalization to larger instances (Altabaa et al., 2025). In pretrained language models, a shared middle block supports depth extrapolation on controlled rule-following tasks, although zero-shot transfer to verbal formulations remains limited (Shapiro, 2026). Looped language models also improve compositional tool calling involving dependent calls (Popescu et al., 2026). **Adaptive Computation.** PoLar (Li et al., 2026b) learns input-dependent programs that skip or repeat groups of layers in a frozen backbone. Think-at-Hard (Fu et al., 2026) selects tokens for extra latent iterations, and its math-only variant also improves on MMLU-STEM. Extra iterations can still degrade predictions when recurrent states drift, limiting depth extrapolation (Viakhirev et al., 2026).

5.2 Information Routing

Information routing determines how tokens are addressed, which dependencies attention can access, and which components process them (Vaswani et al., 2017; Shazeer et al., 2017). For reasoning generalization, these designs aim to preserve the connections needed by a computation as input length or composition changes. **Positional Encoding.** Position Coupling (Cho et al., 2024) assigns shared position identifiers to digits of equal significance across operands, while Abacus embeddings (McLeish et al., 2024) encode digit positions within each number. Structural symmetry and alternative positional representations also improve arithmetic extrapolation (Sabbaghi et al., 2024; Shen et al., 2023). Multi-level position coupling (Cho et al., 2025) combines multiple position coordinates with task-specific scratchpads to generalize across operand lengths and counts. TAPE (Zhu et al., 2025b) instead updates positional representations across layers using sequence context, improving arithmetic length extrapolation. Much of this arithmetic evidence comes from specialized models. **Attention Patterns.** RegularGPT (Chi et al., 2023) combines sliding-dilated attention with shared layers and length-dependent depth, composing local results over larger spans to recognize regular languages beyond training lengths. Attention Bias Calibration (Duan et al., 2023) converts attention patterns learned on shorter inputs into biases that support arithmetic length extrapolation. **Modular Reasoning.** MORSE (Fu and Frank, 2024) learns input-dependent masks that route tokens to specialized attention heads, improving generalization across entailment-tree lengths and shapes. In sparse expert language models, more complex skill compositions benefit from more active experts, while excessive activation can hurt simpler compositions (Zhao et al., 2026b).

5.3 Recurrent Memory

Recurrent memory carries an internal state across computation steps, allowing later steps to reuse earlier information (Bulatov et al., 2022). Updates occur across tokens or reasoning chunks, and generalization depends on retaining the needed information as sequences grow. **Token-Level Recurrence.** Rational Transductors (Mohri, 2026) combine weighted-automaton recurrence with transformer attention, providing theoretical and synthetic-task evidence for length generalization. SST V2 (Aviss, 2026) blends a carried state into pretrained feedforward layers and reports gains beyond math fine-tuning, although its aggregate scores count a question as solved if any tested iteration depth succeeds. **Chunk-Level Recurrence.** Thinking States (Amos et al., 2026) compresses intermediate thoughts into a fixed-size state for later chunks, improving length extrapolation on parity and variable-tracking tasks, although the model still accesses preceding context alongside this state.

6 Analysis of Generalization

While the methods in §3–§5 aim to improve performance under distribution shift, analytical studies examine what transfers and why. We organize this work by three complementary types of evidence: behavioral evidence measures performance under controlled shifts; mechanistic evidence probes internal computation, mostly in small models; and theoretical evidence characterizes what can be represented, learned, and certified. Table 3 and Appendix C summarize each line of analysis.

6.1 Behavioral Evidence

Compositional generalization. Accuracy declines as computation graphs grow deeper and wider (Dziri et al., 2023), models can fail to chain two facts they both know (Zhang et al., 2026b), and this gap does not shrink with scale in the GPT-3 family (Press et al., 2023). **Length, depth, and difficulty.** Transformers fail on nesting deeper than seen in training (He, 2025), short-to-long transfer varies across random seeds (Zhou et al.,

2024d), and training only on easy or only on hard examples yields inconsistent gains (Kordi et al., 2026). **Prior exposure.** Whether a test is "unseen" depends on the training stage it is compared against: accuracy drops on familiar tasks posed in an unfamiliar number base (Wu et al., 2024), facts learned as "A is B" are not retrieved as "B is A" (Berglund et al., 2024), and RL transfer requires sufficient base-model competence (Lu et al., 2025).

6.2 Mechanistic Evidence

Mechanistic studies reveal how internal computations shape generalization. **Shared circuits.** If two problem variants are solved by the same procedure, their internal computations should overlap. de Varda et al. (2026) show that, in LLM arithmetic, the overlap between circuits for numeric and verbal formats predicts which formats are hardest, which models transfer best, and which items are solved. However, such a circuit can transfer poorly once it must use newly integrated knowledge (He et al., 2026). **Interfaces between steps.** Multi-step reasoning can also fail when an intermediate result does not reach the next step. Biran et al. (2024) find that LLMs resolve the bridge entity of a two-hop query in early layers but begin the second hop in later layers that may lack the required knowledge; patching a later hidden state into an earlier layer, chosen per query, recovers up to 66% of failed queries. **Learning dynamics.** When several solutions fit the training data, optimization determines which one the model reaches. Grokked transformers generalize OOD on comparison but not on composition (Wang et al., 2024a). In LLM post-training, OOD performance can peak early in SFT and then decline while ID performance still improves (Jin et al., 2025), making the evaluated checkpoint part of the generalization claim.

6.3 Theoretical Evidence

Theoretical work distinguishes three properties that empirical results often conflate: *expressivity*, whether a model class can represent a procedure that generalizes in length or composition; *learnability*, whether finite training selects this procedure over shortcuts that fit the same data; and *certification*, whether a finite test can establish that it extrapolates. Each result holds only under its stated encoding, precision, and computation budget. **Learnability.** Two conditions govern whether training selects the extrapolating procedure. The first is locality: Abbe et al. (2024) show, theoretically under additional assumptions and empirically, that targets with a high globality degree are not efficiently learnable by regular transformers, whereas inductive scratchpads, which condition each step only on the question and the previous state, break this barrier and improve length transfer. The second is inductive bias: Liu et al. (2023a) prove that $O(\log T)$ layers suffice to simulate any finite automaton on inputs of length T, yet standard training in their experiments finds shortcuts that are brittle outside the training distribution. **Certification.** Yang et al. (2026a) prove that no computable length generalization bound exists for C-RASP, even with two layers, so no computable test length certifies the class. Positive results hold only for narrower classes: one- and two-layer transformers provably length-generalize when behavior on longer sequences can be simulated by behavior on shorter training sequences (Izzo et al., 2026).

7 Challenges and Future Directions

Challenges (1) **Limited transfer.** Models that perform well on familiar tasks can fail when known reasoning steps are combined in new ways or applied to different domains (§6.1). Learning procedures that remain effective under these changes is still a major challenge. (2) **Length and depth extrapolation.** Accuracy can decline when problems require more reasoning steps than those seen during training (Wang et al., 2026c). Additional computation at inference does not always resolve this difficulty, as recurrent states may drift (Viakhirev et al., 2026). (3) **Interpretability.** Behavioral scores alone do not show whether a model reuses the same reasoning procedure across tasks. Many mechanistic studies use small models, and their findings may not fully explain transfer in frontier LLMs (§6.2).

Future Directions (1) **Transferable training.** Developing training methods that teach reusable operations is central to improving generalization. Supervision should emphasize the reasoning steps that remain useful when familiar operations are combined in new ways (§3). (2) **Stable recurrent architectures.** Developing recurrent architectures with stable state updates may improve reasoning on longer and deeper problems. These architectures should maintain reliable computation

as the number of reasoning steps increases (§5). (3) **Interpretability and verification.** Developing methods to identify and verify reusable computations is important for understanding reasoning generalization. Applying these methods to larger models would help connect mechanistic findings with the failures observed under distribution shift (§6.2).

8 Conclusion

This survey reviews reasoning generalization in LLMs from four perspectives: training, inference, architecture, and analysis. Existing studies show that generalization depends on what models learn, the computation available at inference, and the type of distribution shift. Although current methods achieve promising results in specific settings, none consistently generalizes across different shifts. Future research should develop clearer evaluation protocols and better explain when and why reasoning generalizes.

Limitations

While this survey offers a comprehensive review of existing methods and analyses of reasoning generalization in LLMs, a few limitations remain in literature coverage, comparison across studies, and applicability to larger models. Given the broad scope and rapid growth of this field, our review may have missed some relevant studies. Our focus on explicit reasoning-related distribution shifts limits coverage of related work and leaves relatively few inference studies. Differences in models, tasks, metrics, and compute restrict the synthesis to qualitative comparisons. Many mechanistic and theoretical findings rely on small models or idealized assumptions, so their relevance to larger models remains uncertain. Finally, our taxonomy assigns each study a primary pillar, although some methods span several.

Ethical considerations

This work surveys publicly available research on reasoning generalization in LLMs and does not involve human participants, collect new personal data, or train or release new models. Reasoning methods may be used for both beneficial and harmful purposes, and improved transfer on selected benchmarks does not establish that a model is safe or reliable in deployment. We therefore encourage assessing generalization alongside bias, privacy, and misuse risks, especially in applications where reasoning errors can harm users.

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A Literature Collection and Corpus

A.1 Abstract-Based Retrieval

We searched arXiv abstracts for literature available by 22 September 2026, using terms such as "reasoning generalization", "reasoning transfer", "OOD reasoning", "compositional generalization", "systematic generalization", and "length generalization". Supplementary searches used "post-training", "SFT", "GRPO", "RLVR", and "reward" to broaden coverage of training methods. We also checked relevant papers in ACL Anthology and OpenReview.

A.2 Screening and Corpus Construction

After deduplication, we screened abstracts and consulted full papers for studies of reasoning transfer, its mechanisms, or theoretical limits. Empirical transfer claims had to specify the reasoning task, reference exposure, and test shift; benchmark gains alone were insufficient. We excluded unrelated robustness or model-to-model transfer studies, while retaining relevant methods without direct transfer evidence as context. The *corpus version* contains 239 studies across the four pillars, of which 173 form the *survey version*, covering the main text, taxonomy, and appendix evidence. Each paper is counted once under its primary pillar; background, survey, and dataset-only references are excluded. Figure 4 and Table 1 summarize both versions.

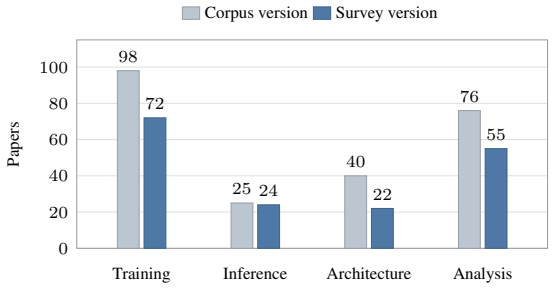


Figure 4: The corpus version ($n = 239$) and author-selected survey version ($n = 173$), by primary pillar. Survey counts include the main text, taxonomy, and appendix evidence; each paper is counted once.

Pillar	Version	≤2023	2024	2025	2026	Total
Training	Corpus	0	5	42	51	98
	Survey	0	4	33	35	72
Inference	Corpus	2	7	9	7	25
	Survey	2	7	9	6	24
Architecture	Corpus	6	7	5	22	40
	Survey	4	6	3	9	22
Analysis	Corpus	13	16	15	32	76
	Survey	10	10	12	23	55
Total	Corpus	21	35	71	112	239
	Survey	16	27	57	73	173

Table 1: Catalogue and survey studies by primary pillar and first-posting year. Venue year is used for records without an arXiv version. Background references and related surveys are excluded from both versions.

B Synthesis and Related Surveys

B.1 Connections Across Pillars

Training and architecture determine which procedures can be reused at inference. Inductive scratchpads improve length transfer by making each step depend only on the question and previous state (Abbe et al., 2024), while recurrent computation requires stable updates as depth increases (Viakhirev et al., 2026). External memory can extend the length range of bounded-state models, but this establishes a capability of the combined system rather than the unaided model (Malach et al., 2026).

Training must also support the inference procedure used at deployment. In controlled shortest-path tasks, test-time scaling remains limited by the reasoning learned during training (Tong et al., 2026). Controller-aware training addresses this mismatch and improves transfer to new inference workflows (Choudhury et al., 2026).

B.2 Comparison with Related Surveys

Table 2 compares representative surveys of reasoning tasks and methods. Their coverage overlaps with ours, but our emphasis is the transfer of reasoning procedures under explicit changes in composition, length, domain, or problem conventions. We connect the methods reviewed with behavioral, mechanistic, and theoretical evidence about the conditions under which they generalize.

C Further Analytical Evidence

This appendix expands each paragraph of §6 under the same heading and adds the expressivity results on which the theoretical paragraphs build. Table 3 summarizes the lines of analysis, while Table 4 lists representative behavioral diagnostics and the shifts they target.

C.1 Behavioral Evidence

Compositional generalization. The decline reported by Dziri et al. (2023) was measured on multiplication, logic-grid puzzles, and dynamic programming. On formal languages, performance depends on whether a prompt stays within the rules learned in training (Mészáros et al., 2024). Accuracy on MathGAP falls as proof trees become deeper, wider, and nonlinear (Opedal et al., 2025), and XDomainBench reports a collapse as more scientific domains are combined, partly because each added domain also makes the problem harder (Gong et al., 2026). Since both benchmarks enlarge the combination together with its depth or difficulty, neither isolates composition; two-hop questions come closer to doing so because the number of steps is fixed. Scale alone does not provide systematicity. On Compositional-ARC, a 5.7M-parameter model meta-trained for compositionality generalizes to unseen combinations of geometric transformations where frontier LLMs do not (Mondorf et al., 2025).

Length, depth, and difficulty. These axes can diverge within a single environment. In a controlled shortest-path setting, models transfer to unseen maps but fail as the planning horizon grows, which the authors attribute to recursive instability (Tong et al., 2026). Length transfer is usually partial. Scratchpads improve extrapolation on algorithmic tasks without removing the underlying failure (Anil et al., 2022), and the seed dependence reported by Zhou et al. (2024d) is consistent with underspecification: the same data admit both an extrapolating solution and a shortcut, and each training run settles on one of them (§6.3). Length generalization can also transfer between tasks, as training on a longer, related auxiliary task helps a target task extrapolate beyond its own training lengths (Cai et al., 2025). For difficulty, ordering deductive-reasoning data from easy to hard gives no robust advantage over random order under either SFT or RL (Mordig et al., 2026).

Prior exposure. In chess, both performance and the benefit of each additional reasoning token decline as relevant priors become sparse (Pleiss et al., 2026). When a hard perturbation invalidates the original solution, models keep applying the mem-

Survey	Main focus
Latent reasoning (Chen et al., 2026c; Zhu et al., 2025c)	Latent representations, horizontal and vertical recurrence, training, and internal mechanisms
Logical reasoning (Cheng et al., 2025a)	Logical question answering and consistency, including solver-based, prompting, and fine-tuning methods
Automated planning (Aghzal et al., 2025)	LLMs as planners or components of planning systems, with emphasis on long horizons
Inductive reasoning (Chen et al., 2025b)	Inferring rules from observations through post-training, test-time exploration, and data augmentation
Efficient reasoning (Wang et al., 2026b)	Reducing reasoning cost through post-training, CoT compression, and adaptive test-time computation
Mathematical reasoning (Wang et al., 2025)	Mathematical comprehension and answer generation through pre-training, prompting, SFT, RL, and extended CoT
Multimodal reasoning (Li et al., 2025)	Development from perception-driven pipelines to multimodal CoT, RL, and native reasoning models
Neuro-symbolic reasoning (Yang et al., 2025)	Symbolic supervision, LLM-assisted symbolic reasoning, and hybrid neural-symbolic systems
Reasoning generalization (ours)	Transfer of reasoning procedures across distribution shifts, covering training, inference, architecture, and behavioral, mechanistic, and theoretical evidence.

Table 2: Main focus of related reasoning surveys and this work.

orized technique, and they do so more often when the original problem appears in context (Huang et al., 2025a). Tasks designed to fall outside pre-training, such as programming in esoteric languages, reveal failures that familiar benchmarks hide (Sharma and Chopra, 2026). Knowledge that is present is not always used. Models that cannot reverse a fine-tuned fact succeed when the same fact is given in context (Berglund et al., 2024), and models can acquire abstract structure yet make only limited use of it for compositional generation at test time (Chen et al., 2026b).

Key takeaways

- **Transfer is axis-specific.** Success on one named shift does not predict success on another.
- **Knowing is not composing.** Stored components can fail to combine, even at scale.
- **"Unseen" is stage-relative.** Apparent generality depends on the reference-exposure stage.

C.2 Mechanistic Evidence

Shared circuits. de Varda et al. (2026) localize their circuits with attribution patching. In small transformers, causal interventions can recover compact RASP subprograms that suffice to explain length-generalizing models (Huang et al., 2026), so reusable computation can be located directly instead of being inferred from scores. Reuse has identifiable limits. Whether pattern matching succeeds on an instance is predicted by how many training contexts witness the relevant functional equivalence, and when a variable reaches the output through several paths, models fail to form a single intermediate representation (Chang et al., 2025). Together with the failure on newly integrated knowledge noted in §6.2 (He et al., 2026), these limits mean that overlap measured on familiar data supports a transfer claim only if it also holds on the shifted split.

Interfaces between steps. In transformers trained from scratch on two-hop queries, lower layers build a representation of the bridge entity, but when the second hop departs from its training distribution, upper layers map that representation to outputs instead of reasoning over it (Zhang et al., 2026b). For facts injected by fine-tuning, relocating their internal representations by self-patching substantially improves failed cases, which is consistent with the facts being stored but not routed to the layers where later computation uses them (Dai et al., 2026). In small transformers trained from scratch on SCAN, a homomorphism error measured on hidden states predicts compositional transfer under noise injection, and regularizing this error during training improves the transfer (An and Du, 2026). Because the regularizer is a training intervention, this result goes beyond correlation, although it may change more than the measured property. Read alongside the two-hop results in §6.2, these findings locate the bottleneck at the hand-off between operations rather than in the presence of the facts or subskills.

Learning dynamics. With extended training, vanilla transformers come to generalize hierarchically on structurally novel inputs long after in-domain accuracy has saturated (Murty et al., 2023). A grokked rule can nevertheless remain bound to the facts seen in training (Wang et al., 2024a), which mirrors the limit of circuit reuse on new knowledge. Regularization can act within a critical period: on a controlled compositional task, weight decay applied during a single window of training matches the out-of-distribution accuracy of weight decay applied throughout training (Ali, 2026). In LLM post-training, excessive SFT can erode the plasticity that later RL needs (§3.3.3; Liu et al., 2026c). Each of these findings is tied to particular models, seeds, and finite test ranges, and theory is needed to separate such regularities from guarantees.

Key takeaways

- **Overlap predicts transfer.** Shared circuits predict transfer across variants, yet can fail on new knowledge.
- **Failures sit at interfaces.** Computed intermediate results may not reach the next step.
- **Optimization selects the solution.** Training length and the evaluated checkpoint shape what transfers.

C.3 Theoretical Evidence

Expressivity. Learnability results presuppose that the model class can represent the target procedure, and what it can represent depends on the computation budget and the state encoding. The number of intermediate steps is itself a resource. Under assumptions on layer normalization, transformer decoders with linearly many decoding steps recognize all regular languages, and with polynomially many steps they recognize exactly the problems solvable in polynomial time (Merrill and Sabharwal, 2024). Without intermediate steps, constant-depth transformers with constant-bit precision are limited to AC^0; with as many chain-of-thought steps as the size of a Boolean circuit, they can evaluate that circuit (Li et al., 2024c). Extrapolating constructions further depend on how state is encoded. A universal-transformer construction provably evaluates Boolean circuits of any length and depth, but it assumes fully parenthesized inputs, a positional encoding that tracks gate depth, and a number of iterations that grows with depth (Ito et al., 2026). Such constructions do not carry over directly to finite-precision LLMs, and representability alone does not show that training will find them.

Learnability. Scratchpads whose content is unsupervised do not break the globality barrier of Abbe et al. (2024), and the vocabulary of the trace matters as well. Under standard positional encodings and a finite alphabet, chain-of-thought learning that generalizes in length cannot go beyond TC^0, whereas signpost tokens, whose vocabulary grows with problem size, allow a length-generalizable simulation of Turing machines (Kraus et al., 2026). Decomposition can reduce sample complexity. When the learner receives feedback on intermediate states, a recursive curriculum for simulating semiautomata over T steps reduces the required supervision from $\Omega(T)$ to $2^{O(\sqrt{\log T})}$ tokens (Rajaraman et al., 2026). Empirically, ablations in tiny transformers trained on Dyck languages attribute the gain from intermediate traces to their content rather than to the extra tokens (Valle and Reid, 2026). These results suggest why scratchpads give only partial gains under length shift: a trace helps when it turns a global dependency into local, reusable updates, and otherwise it adds tokens without changing what must be learned.

Other results concern inductive bias, which determines which of several fitting solutions training reaches. Beyond the logarithmic-depth construction in §6.3, Liu et al. (2023a) show that solvable semiautomata can be simulated with depth independent of T. Zhou et al. (2024a) conjecture, with experimental support, that transformers length-generalize on tasks that have a short program in RASP-L, a restricted language of transformer-compatible operations. On synthetic state-tracking tasks, Huang et al. (2025c) prove that gradient descent learns chains of thought whose length extrapolation is governed by the algebraic structure of the task, through attention concentration. Kazemnejad et al. (2023) prove that a decoder without positional encoding can represent both absolute and relative positions, and they observe that after training its attention mostly resembles a relative scheme and extrapolates better than explicit encodings. In the retrieval task of Singh et al. (2026), rotary encodings make attention logits depend only on the relative offset, whereas a head with learned absolute encodings stays fixed to a position inside the training range. Data matter as well, but they

are not sufficient on their own: the guarantees of Ahuja and Mansouri (2024) for length and compositional transfer require both diverse training support and constraints that rule out direct memorization.

Certification. The negative result in §6.3 concerns C-RASP, a formalism linked to length generalization, so its reach depends on what C-RASP can express; for regular languages, this is determined by their algebraic decomposition (Yang et al., 2026b). For the positive fragment of C-RASP, which is equivalent to fixed-precision transformers, a computable length generalization bound exists but is necessarily exponential (Yang et al., 2026a). For causal transformers with learnable absolute positional encodings, Huang et al. (2025b) characterize the functions that an idealized inference scheme with a norm-based regularizer identifies in the limit. This characterization proves that length generalization is possible for a rich family of problems, and in experiments it predicts success and failure across algorithmic and formal-language tasks. The positive result of Izzo et al. (2026) in §6.3 also bounds the training length it requires. Plan verification provides a positive case for checking rather than generating: transformers provably learn to verify long plans in a large class of classical planning domains, even as the vocabulary grows with the number of objects (Sarrof et al., 2026).

Key takeaways

- **Representable is not learnable.** Expressible procedures need not be learned.
- **Locality and bias govern selection.** Global targets resist learning without local steps, and standard training can prefer brittle shortcuts.
- **Finite tests do not certify.** Accuracy on a finite range is evidence about that range alone.

D Datasets and Evaluation Metrics

This appendix describes the resources behind the evidence in the main text: the supervision a training dataset provides, the shift an evaluation set implements, and the quantity a reported score measures. Tables 6 and 10 list the datasets, and Appendix D.3 documents the comparison of hybrid methods in §3.3.3.

D.1 Datasets

D.1.1 Datasets for Training

Supervision. Training datasets differ in the supervision they provide. Prompts with verifiable answers support outcome-reward training without demonstrated solutions: DAPO-Math rewrites problems to obtain integer answers, and DeepScaleR, ORZ-Math, and Big-Math enlarge the pool of verifiable prompts (Yu et al., 2025; Luo et al., 2025; Hu et al., 2025a; Albalak et al., 2025). GSM8K and MATH include human-written solutions and can support either supervised or reward-based training (Cobbe et al., 2021; Hendrycks et al., 2021). Demonstration corpora supply reasoning traces to imitate. NuminaMath-CoT contains reformatted solutions, OpenR1 pairs NuminaMath 1.5 problems with several DeepSeek-R1 traces (Li et al., 2024a; Hugging Face, 2025), and LIMO and s1K select small sets of 817 and 1,000 demonstrations, respectively (Ye et al., 2025; Muennighoff et al., 2025). These categories describe the supervision a dataset offers rather than its only possible use: a trace corpus can supply prompts for RL, and a generator can produce supervised traces.

Controllable distributions. Knights and Knaves, Reasoning Gym, SynLogic, and Enigmata provide generators and verifiers with adjustable task parameters (Xie et al., 2025; Stojanovski et al., 2025; Liu et al., 2025a; Chen et al., 2025a), so training and test sets can be separated deliberately by length, structure, or difficulty. A held-out random seed alone yields new instances, not a structural shift. Decontamination serves a different purpose: removing benchmark overlap, as in DeepMath, reduces leakage but does not define an OOD split (He et al., 2025b). Multi-domain resources such as GURU and WebInstruct-verified support source-to-target studies, and DataFlex-RL fixes a mixture of math, logic, and science data to compare data policies (Cheng et al., 2025c; Ma et al., 2025b; Liang et al., 2026). For these resources, domain coverage and verifier reliability should be reported separately.

Counting units. Table 6 counts problems unless stated otherwise, because problems, traces, and task types are not comparable units. OpenR1-Math-220k, for example, contains 225,129 problems in its full release (93,733 in the default configuration and 131,396 in the extended one), each

Line of analysis	Question answered	Main finding	Interpretation limit
Behavioral evidence (§6.1)			
Compositional generalization	Do familiar primitives transfer to new combinations?	Loss grows with the depth, width, and order of the combination	Template overlap can make recombination look like composition
Length, depth, and difficulty	Does the procedure continue beyond the trained range?	Transfer is partial, axis-specific, and sensitive to the training seed	A finite extrapolation range is not unbounded extrapolation
Prior exposure	How far does the test lie from earlier training exposure?	Benefits shrink as priors become sparse, conventions change, or facts are queried in the unseen direction	Pre-training exposure is inferred, not observed
Mechanistic evidence (§6.2)			
Shared circuits	Do successes on both sides of a shift share a computation?	Circuit overlap predicts format transfer and item-level correctness	Overlap on familiar data need not hold on the shifted split
Interfaces between steps	Can each step read the result of the previous one?	Bridge entities are bypassed, resolved too late, or stored out of reach; homomorphism error predicts transfer	Repairing patches are selected per case; training interventions may change more than the measured property
Learning dynamics	Which of several fitting solutions does training select?	Rules emerge late, can stay bound to seen facts, and depend on regularization timing	Tied to specific models, seeds, and checkpoints
Theoretical evidence (§6.3)			
Learnability	Will training on finite data select a procedure the model class can represent?	Training finds brittle shortcuts unless locality, diverse support, or symmetry favor extrapolating solutions	Representability does not imply learnability; assumptions rarely match LLM training
Certification	Can a finite test establish extrapolation?	Finite training lengths suffice for restricted classes; no computable test length covers all two-layer C-RASP programs	Positive results assume idealized training or cover verification only

Table 3: Lines of analysis: the question each answers, its main finding, and the limit on interpreting it. Rows follow the paragraph headings of §6. Behavioral rows rest on performance under a named shift, mechanistic rows on representational or interventional evidence, and theoretical rows on results under explicit assumptions.

with two to four traces (Hugging Face, 2025). The hybrid comparisons in §3.3.3 use OpenR1-Math-46k-8192, a subset of the default configuration with correct, short traces (Yan et al., 2025; Limozin et al., 2026).

D.1.2 Datasets for Evaluation

Table 5 summarizes the main distribution shifts used in reasoning evaluations.

Held-out benchmarks. Mathematical post-training is commonly evaluated on AIME, AMC, MATH-500, Minerva/OCW, and OlympiadBench, with ARC-Challenge, GPQA-Diamond, and MMLU-Pro as checks outside mathematics (Yan et al., 2025; Limozin et al., 2026). We refer to these as the mathematics and out-of-domain suites, although even the mathematical tests may differ from the training data in difficulty, subject, or source. SuperGPQA, HLE, ZebraLogic, BBH, HumanEval, and LiveCodeBench broaden the range of content (Du et al., 2025; Phan et al., 2026; Lin et al., 2025; Suzgun et al., 2023; Chen et al., 2021; Jain et al., 2024). Whether a benchmark is ID or OOD depends on the model's pre-training, its adaptation data, and the task definition, so the label cannot be assigned to the benchmark alone.

Controlled shifts. Controlled benchmarks aim to change one property while holding the others fixed. GSM-IC adds irrelevant context, GSM-Plus applies perturbations that may also change difficulty, and GSM-Symbolic samples variants from mathematical templates (Shi et al., 2023; Li et al., 2024b; Mirzadeh et al., 2025). SCAN, COGS, and CFQ hold out lengths or combinations of familiar primitives (Lake and Baroni, 2018; Kim and Linzen, 2020; Keysers et al., 2020). MathGAP varies proof depth and width, Compositional-ARC holds out geometric compositions, and OMEGA separates greater complexity from new combinations and new strategies (Opedal et al., 2025; Mondorf et al., 2025; Sun et al., 2025). CLRS-30 was developed to probe neural algorithmic executors and is not a standard LLM leaderboard (Veličković

Diagnostic	Primary shift	Main observation	Interpretation limit
GSM-Plus (Li et al., 2024b)	Statements and answer targets	Correctness is brittle even on variants of solved problems	Perturbations remain within grade-school arithmetic
GSM-Symbolic (Mirzadeh et al., 2025)	Numeric instantiation and distractor clauses	Equivalent templates yield unstable accuracy	Template generation cannot test every skill
MathGAP (Opedal et al., 2025)	Proof-tree depth, width, and shape	Accuracy declines with structural complexity	Generated word problems simplify language variation
Compositional-ARC (Mondorf et al., 2025)	Unseen transformation combinations	Meta-learned small models can exceed frontier LLM systematicity	Abstract spatial tasks are controlled but narrow
General365 (Liu et al., 2026a)	Domain and variant	Broad reasoning remains below specialized math performance	Variant construction may not remove familiarity
XDomainBench (Gong et al., 2026)	Cross-domain composition	Failure grows with composition order and interaction	Knowledge and reasoning are partly entangled
Numeric remapping (Barker et al., 2026)	Surface-preserving numeric change	Robustness differs sharply by dataset structure	Arithmetic word problems only
Elenchos (Steiglechner et al., 2026)	Abductive rule mutation	Detection is stronger than causal attribution	Formal systems are controlled but narrow
OOD visual planning (Neuhaus et al., 2026)	Map size and representation	CoT helps ID more than most OOD settings	Simple navigation does not span open-world planning
Chess generalization (Pleiss et al., 2026)	Density of useful priors	Performance falls as relevant priors become sparse	Training exposure is inferred, not observed
OMEGA (Sun et al., 2025)	Mathematical novelty type	Separates exploratory, compositional, and transformative transfer	Novelty taxonomy depends on task construction
CoT as a mirage (Zhao et al., 2025)	Data distribution	Apparent reasoning depends on distribution design	Behavioral evidence does not identify a mechanism

Table 4: Representative behavioral diagnostics: the shift each targets, its main observation, and the limit on interpreting it.

et al., 2022). EsoLang-Bench changes the programming language while keeping the algorithmic problems; the scarcity of public data in these languages is a proxy for low exposure, not proof that pre-training never included them (Sharma and Chopra, 2026). General365 derives 1,095 variants from 365 seed problems, so statistical analyses should account for the shared seeds (Liu et al., 2026a).

Recency and contamination. Examination items and rolling benchmarks support temporal checks only when the items postdate the evaluated model's relevant training cutoff (Jain et al., 2024; White et al., 2024); AIME 2025, for instance, is not automatically unseen for models trained after its release. GSM1K was withheld at its initial publication, which supported its original comparison with GSM8K, but evaluations after its public release require a new exposure audit (Zhang et al., 2024b). Neither a publication date nor exact-match decontamination excludes exposure to equivalent problems. Table 10 therefore reports the evaluation axis and dataset version of each benchmark instead of a fixed OOD label.

D.2 Evaluation Metrics for Reasoning Generalization

Reliability under a fixed decoding protocol. For N problems and a task-specific correctness checker V_q, single-sample accuracy is

$$\mathrm{Acc} = \frac{1}{N}\sum_{q=1}^{N} V_q(\hat{y}_q).$$

The checker may normalize mathematical expressions, execute code, or compare labels. With k independently sampled responses per problem,

$$\mathrm{avg@}k = \frac{1}{N}\sum_{q}\frac{1}{k}\sum_{s=1}^{k} V_q(\hat{y}_{q,s}).$$

This quantity estimates the expected single-sample accuracy under the chosen decoding settings; it does not measure success within k attempts. Maj@k instead grades the most frequent normalized answer under a stated tie-breaking rule (Wang et al., 2023), so it evaluates an aggregation procedure that spends k samples at deployment.

Coverage at a stated budget. Pass@k measures whether at least one of k samples succeeds. Given $n \geq k$ independent samples at fixed decoding

Shift axis	Train → test	What it probes
Instance	new samples, same generator	statistical generalization
Surface / format	paraphrase, renamed symbols, new notation	invariance to irrelevant form
Composition	new combinations of seen primitives	rule reuse and recombination
Length / depth	longer inputs, more hops, deeper nesting	extrapolation of iteration
Difficulty	easier or harder than training range	transfer across capability regimes
Task / domain	new task family or subject	portability of procedures
Language / modality	other language, format, image, audio	reasoning vs. interface
Environment / tool	new APIs, dynamics, rewards	agentic and procedural transfer
Knowledge	new facts, familiar rules	rule use on new content

Table 5: Axes along which reasoning evaluations shift between training and test. When several axes change together, the source of an improvement or failure becomes harder to identify.

settings, of which c_q are correct, the estimator is (Chen et al., 2021)

$$\text{pass@}k = \frac{1}{N}\sum_q \left[1 - \frac{\binom{n-c_q}{k}}{\binom{n}{k}}\right],$$

with $\binom{n-c_q}{k} = 0$ when $n - c_q < k$. Because pass@k relies on an oracle success criterion, it does not measure the accuracy of a deployable selector unless a suitable verifier is available. A higher pass@1 with unchanged or lower large-k coverage is consistent with post-training that reweights existing solutions (Yue et al., 2025), although finite sampling cannot establish with certainty whether a solution is present or absent. We therefore recommend reporting coverage curves together with sample and token budgets.

Robustness to a matched transformation. For comparable reference and shifted sets, both scores should be reported together with the absolute and relative drops,

$$\Delta = \text{Acc}_{\text{ref}} - \text{Acc}_{\text{shift}},$$
$$\Delta_{\text{rel}} = \Delta / \text{Acc}_{\text{ref}},$$

where the relative drop requires positive reference accuracy. Under the same 0–1 loss and evaluation procedure, Δ equals the transfer gap defined in §2. A smaller gap is not necessarily better, since it may result from lower reference accuracy. When variants share an original problem or template, results are best aggregated at that level. GSM-Symbolic reports variation across generated test sets (Mirzadeh et al., 2025), and GSM-IC also reports the fraction of base problems solved under *all* distractor variants (Shi et al., 2023). Conditional variant accuracy, $P(\text{variant correct} \mid \text{original correct})$, measures how often correctness is retained on the originally solved subset, but it is not a complete measure of consistency.

Length and composition. We recommend reporting length results as curves that extend beyond the training frontier and that distinguish input length, required depth, and the budget of generated tokens. As a summary, we define the largest tested length up to which accuracy stays at or above a threshold τ:

$$L^*_\tau = \max\left\{L \in \mathcal{L} : \min_{\ell \in \mathcal{L},\, \ell \le L} \text{Acc}(\ell) \ge \tau\right\}.$$

Here $\mathcal{L}$ is the declared grid of test lengths, and an empty set means that no tested length reaches the threshold. The summary says nothing about untested lengths. The ratio $L^*_\tau / L_{\text{train}}$ describes extrapolation relative to the final training frontier, similar to the length ratios reported in integer-addition studies (Zhou et al., 2024d). For composition, held-out structures should be compared at matched depth or length where possible (Park et al., 2025).

Transfer and retention. Transfer should be measured against the base model on each target task. For method m trained on source s, we define

$$T^{(m)}_{s\to t} = \text{Acc}_t(\theta^{(m)}_s) - \text{Acc}_t(\theta_0).$$

A positive value on a target excluded from adaptation supports transfer, and a negative value records degradation. Sequential retention is a different quantity: it compares an earlier and a later checkpoint on domains that were already learned. Raw accuracy differences across domains also reflect task difficulty and should not be read as a normalized measure of transfer.

Dataset	Year	Domain	Problems / Tasks	Supervision
Math: Verifiable Prompts for RL				
GSM8K (train) (Cobbe et al., 2021)	2021	Grade-school math	7,473	Answer + rationale
MATH (train) (Hendrycks et al., 2021)	2021	Competition math	7,500	Answer + solution
SimpleRL-Zoo (Zeng et al., 2025)	2025	Math	∼8K per split	Answer
DAPO-Math-17k (Yu et al., 2025)	2025	Competition math	∼17K	Integer answer
DeepScaleR (Luo et al., 2025)	2025	Competition math	∼40K	Answer
DeepMath-103K (He et al., 2025b)	2025	Math	103K	Answer + traces
Skywork-OR1 (He et al., 2025a)	2025	Math / code	∼105K / 13.7K	Answer / tests
ORZ-Math (Hu et al., 2025a)	2025	Competition math	∼129K	Answer
Big-Math (Albalak et al., 2025)	2025	Math	>250K	Answer
Math: Demonstrations for SFT, Distillation, and Hybrid Training				
NuminaMath-CoT (Li et al., 2024a)	2024	Math	∼860K	CoT solution
OpenR1-Math-220k (all) (Hugging Face, 2025)	2025	Math	∼225K	2–4 R1 traces per problem
OpenR1-Math-46k-8192†§ (Yan et al., 2025)	2025	Math	∼46K	Long CoT (R1) + answer
LIMO (released set) (Ye et al., 2025; GAIR, 2025)	2025	Math	817	Long CoT
s1K† (Muennighoff et al., 2025)	2025	Math, science	1,000	Long CoT
OpenThoughts releases† (Guha et al., 2025)	2025	Math, code, science	114K / 1M / 1.2M	Long CoT
Logic, Puzzles, and Procedural Environments				
Knights & Knaves‡ (Xie et al., 2025)	2025	Logic puzzles	Procedural	Assignment checker
Reasoning Gym‡ (Stojanovski et al., 2025)	2025	Algorithmic, logic	100+ generators	Task verifier
SynLogic‡ (Liu et al., 2025a)	2025	Logic	35 task types	Task verifier
Enigmata‡ (Chen et al., 2025a)	2025	Puzzles	36 task types	Task verifier
Multi-domain				
SciQ (train) (Welbl et al., 2017)	2017	Science MCQ	11,679	Correct option
GURU (Cheng et al., 2025c)	2025	Six domains	92K	Domain verifiers
WebInstruct-verified (Ma et al., 2025b)	2025	Multi-domain	∼230K	Model-based verifier
DataFlex-RL mixture (Liang et al., 2026)	2026	Math, logic, science	15K	Rule-based checks

Table 6: Summary of datasets used to train reasoning models. † Counts refer to problems, not traces; a problem may have several traces. ‡ Procedural generator; task counts are not instance counts. § Correct, short-trace subset of the OpenR1 default configuration. "∼" marks rounded counts.

Aggregation, uncertainty, and cost. A domain-balanced summary averages within domains before averaging across them:

$$S_{\mathrm{DB}} = \frac{1}{K}\sum_{d=1}^{K}\frac{1}{|\mathcal{B}_d|}\sum_{b\in\mathcal{B}_d}\mathrm{Acc}_b.$$

This average assumes comparable units, so rubric credit, F1, and accuracy should not be averaged without a justified normalization. Benchmark coverage alone can reverse method rankings (Liang et al., 2026), which is why per-task scores should accompany any aggregate. Paired method differences need uncertainty estimates over matched training seeds; item-level bootstrap tests, such as those reported for LUFFY, do not capture training-seed variance (Yan et al., 2025). Bootstrap resampling of generated variants should operate on their shared problem or template. Run-level standard deviations and confidence intervals answer different questions and need to be labeled accordingly, and inference methods are compared at a fixed budget or through accuracy-cost curves (Snell et al., 2024).

D.3 Protocols and Published Results

Common reproduction study. Table 7 is taken from Limozin et al. (2026), in which the corrected Qwen2.5-Math-7B (Yang et al., 2024) baselines and the LUFFY and ReLIFT reproductions all use OpenR1-Math-46k-8192 (Hugging Face, 2025; Yan et al., 2025). AIME24, AIME25, and AMC are scored with avg@32 and the remaining tests with pass@1, using a temperature of 0.6, a maximum response length of 8,192, and shuffled options for multiple-choice questions. The corrected baselines come from three independent end-to-end runs, whereas each reproduction is a single run. Sequential training consists of SFT followed by 500 GRPO steps (Shao et al., 2024), and the 50-step variant also raises the RL learning rate from 10^{-6} to 5×10^{-6}. The comparison therefore shares an evaluation protocol but does not match

compute.

Published results. Table 8 keeps the originally reported hybrid results separate from the reproduction, and Table 9 summarizes how each method combines SFT and RL. All five methods build on GRPO-family objectives: LUFFY adds off-policy R1 traces to each GRPO rollout group, ReLIFT interleaves GRPO with SFT on questions that all rollouts fail, SRFT adds an entropy-weighted SFT loss to group-relative RL losses, Prefix-RFT applies Dr. GRPO (Liu et al., 2025c) to rollouts that continue demonstration prefixes, and HPT switches each question between SFT and a Dr. GRPO-style RL loss according to rollout accuracy. LUFFY and Prefix-RFT report the full nine-benchmark suite (Yan et al., 2025; Huang et al., 2025d). ReLIFT uses avg@8 on MATH-500 and OlympiadBench and does not report Minerva (Ma et al., 2025a), SRFT omits AIME25 (Fu et al., 2025), and HPT reports only two OOD tests, omitting MMLU-Pro, with top-p sampling (Lv et al., 2025). We leave these cells empty rather than importing values from later compiled tables. The original comparisons also differ in SFT implementation and hyperparameters, as audited by Limozin et al. (2026), so the gap between published and reproduced scores mixes implementation differences with seed variance.

Method	Math avg.	ARC-C	GPQA-D	MMLU-Pro	OOD avg.
Corrected SFT	52.2 ± 0.2	82.5 ± 0.6	24.6 ± 2.9	50.2 ± 0.3	52.4 ± 1.1
SFT→GRPO (50 steps)	55.6 ± 0.5	84.0 ± 0.5	39.6 ± 1.0	53.9 ± 0.4	59.2 ± 0.2
SFT→GRPO (500 steps)	57.0 ± 0.3	84.0 ± 0.7	40.6 ± 1.8	55.1 ± 0.6	59.9 ± 0.9
LUFFY	46.3	81.4	41.8	50.9	58.0
ReLIFT	48.8	80.7	36.7	51.3	56.2

Table 7: Accuracy (%) from Limozin et al. (2026); the originally published results are given in Table 8. All rows use Qwen2.5-Math-7B (Yang et al., 2024) and OpenR1-Math-46k-8192 (Hugging Face, 2025; Yan et al., 2025). Corrected baselines report mean ± standard deviation over three independent runs; mixed-policy reproductions use one run. Math avg. covers six benchmarks and OOD avg. covers the three displayed benchmarks. Training compute is not matched across all rows.

Method	AIME24	AIME25	AMC	MATH-500	Minerva	Olympiad	Math avg.	ARC-C	GPQA-D	MMLU-Pro	OOD avg.
LUFFY (Yan et al., 2025)	29.4	23.1	65.6	87.6	37.5	57.2	50.1	80.5	39.9	53.0	57.8
ReLIFT (Ma et al., 2025a)	28.3	22.9	65.1	87.9	—	57.3	—	81.6	43.1	53.9	59.5
SRFT (Fu et al., 2025)	35.3	—	74.3	89.8	39.7	58.3	—	85.3	46.4	55.9	62.5
Prefix-RFT (Huang et al., 2025d)	31.8	26.4	68.2	88.4	40.3	55.7	51.8	84.0	39.1	52.1	58.4
HPT (Lv et al., 2025)	33.0	21.9	69.4	89.2	46.0	56.9	52.7	81.6	42.9	—	—

Table 8: Originally reported hybrid scores (%). A shared model and nominal dataset do not imply identical evaluation or compute. ReLIFT uses avg@8 on two mathematical tests; SRFT and HPT omit different tests. Dashes denote absent original results. Means cover the complete six-math or three-OOD suite; ReLIFT's three-test OOD mean is calculated here from its Tables 1 and 5.

Method	RL algorithm	How SFT and RL are combined	Evaluation or comparison limit
Corrected SFT	None	SFT on demonstrations only; corrected implementation	Three independent runs
SFT→GRPO (50 steps)	GRPO	Full SFT, then 50 GRPO steps	Higher RL learning rate; three runs
SFT→GRPO (500 steps)	GRPO	Full SFT, then 500 GRPO steps	Three runs; extra SFT compute
LUFFY (Yan et al., 2025)	Mixed-policy GRPO	Off-policy R1 traces added to each on-policy rollout group; policy shaping of the off-policy ratio	Published or one-run reproduced results, kept separate
ReLIFT (Ma et al., 2025a)	GRPO†	GRPO interleaved with SFT steps on questions whose rollouts all fail	Original avg@8 on two tests; no Minerva
SRFT (Fu et al., 2025)	GRPO-style, LUFFY-like off-policy term	One stage: entropy-weighted SFT loss plus group-relative RL losses on demonstrations and own rollouts	Original table omits AIME25
Prefix-RFT (Huang et al., 2025d)	Dr. GRPO	One of eight rollouts continues a demonstration prefix; only high-entropy prefix tokens are updated	Original reported results
HPT (Lv et al., 2025)	Dr. GRPO-style GRPO‡	Per-question gate: SFT on the demonstration if all eight rollouts fail, RL otherwise	Two OOD tests; top-p sampling

Table 9: Settings and limits of the hybrid comparison, including the RL algorithm of each method (GRPO: Shao et al., 2024; Dr. GRPO: Liu et al., 2025c). All listed methods use Qwen2.5-Math-7B (Yang et al., 2024) and the filtered OpenR1 data (Hugging Face, 2025; Yan et al., 2025), but training allocations and evaluation details vary. †Without length and standard-deviation normalization, as in Dr. GRPO. ‡The setup names GRPO and the loss definition specifies Dr. GRPO: responses are not length-normalized, but advantages keep standard-deviation normalization.

Dataset / Benchmark	Characteristics and Evaluation Use	Dataset Size / Subset	Evaluation Metric
Mathematics			
GSM8K (Cobbe et al., 2021)	Grade-school word problems requiring multi-step arithmetic.	7,473 train / 1,319 test	Accuracy
MAWPS (Koncel-Kedziorski et al., 2016)	A repository of arithmetic and algebraic word problems.	3,320 problems	Protocol-dependent
Minerva / OCW (Lewkowycz et al., 2022)	Undergraduate STEM problems from MIT OpenCourseWare, including numeric and symbolic answers.	272 problems	Accuracy
MATH (Hendrycks et al., 2021)	Competition-level mathematics problems spanning multiple subjects and difficulty levels.	12,500 problems (7,500 train / 5,000 test)	Accuracy (exact match on the final answer)
MATH-500 (Lightman et al., 2024)	A held-out subset of MATH used to evaluate mathematical reasoning without overlap with the expanded MATH training split.	500 problems	Accuracy; pass@k when reported
AMC	American Mathematics Competitions problems. The number of questions depends on the selected years and evaluation protocol; one released protocol contains 83 integer-answer problems from AMC12 2022–2023.	Protocol-dependent	Accuracy; pass@k when reported
AIME 2024 / 2025 / 2026	American Invitational Mathematics Examination problems. Each edition contains 15 questions; combining AIME I and II yields 30 questions per year.	15 per edition; 30 per year when AIME I and II are combined	Accuracy (exact match); pass@k when reported
CMIMC 2025; BRUMO 2025	Recent university-hosted mathematics competitions used in some evaluation protocols.	Protocol-dependent	Protocol-dependent
OlympiadBench (He et al., 2024a)	A bilingual, multimodal benchmark of Olympiad-level mathematics and physics problems, with expert annotations for step-by-step reasoning.	8,476 problems	Accuracy
MATH^2 (Shah et al., 2024)	A collection of difficult mathematics questions generated with AI assistance and refined through human involvement.	210 problems	Accuracy
ProcessBench (Zheng et al., 2025)	Evaluates process-level reasoning by asking models to identify the earliest erroneous step in a solution or determine that all steps are correct. Its difficulty spans grade-school to competition- and Olympiad-level mathematics.	3,400 test cases	F1 and ArithACC, as reported by the evaluation protocol
Science, Knowledge, and Factuality			
ARC (Challenge / Easy) (Clark et al., 2018)	Grade-school multiple-choice science questions. ARC-Challenge is the harder split, while ARC-Easy contains questions that baseline retrieval and word-co-occurrence methods can answer.	ARC-Challenge: 1,119 train / 299 validation / 1,172 test	Accuracy
GPQA-Diamond (Rein et al., 2024)	Expert-written graduate-level questions in biology, physics, and chemistry. Diamond retains questions answered correctly by both domain experts but missed by most non-experts.	198 questions	Accuracy
MMLU-Pro (Wang et al., 2024b)	Broad-subject multiple-choice benchmark with more reasoning-focused questions and up to ten answer options, compared with four in MMLU.	12,032 questions	Accuracy
SuperGPQA (Du et al., 2025)	Graduate-level knowledge and reasoning questions spanning a broad range of specialized disciplines.	26,529 questions	Accuracy
Humanity's Last Exam (Phan et al., 2026)	Expert-level, closed-ended questions spanning diverse academic fields, designed to evaluate advanced model capabilities.	2,500 questions (public release)	Accuracy
ScienceArena (Zhao et al., 2026a)	Questions drawn from public scientific olympiad competitions in physics, chemistry, and biology; evaluation uses process-credit rubrics.	13 competitions	Rubric-based scoring

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Table 10 (continued)

Dataset / Benchmark	Characteristics and Evaluation Use	Dataset Size / Subset	Evaluation Metric
C-Eval Hard (Huang et al., 2023)	Hard subset of C-Eval, a Chinese-language multiple-choice evaluation suite covering a broad range of academic subjects.	Protocol-dependent	Accuracy
FACTOR (Muhlgay et al., 2024)	Factuality benchmarks generated from a text corpus by perturbing facts; the model should prefer the factual completion. Wiki- and News-based variants exist.	Generated; per-variant sizes in the source	Accuracy
HaluEval-Summary	Summarization subset of HaluEval for detecting hallucinated summaries; used in evaluation of divide-and-conquer prompting (Zhang et al., 2024c).	Protocol-dependent	Accuracy; F1; Precision; Recall
SciFact (misinformation variant)	Author-constructed paragraph-level misinformation variant of SciFact; results depend on the construction in Zhang et al. (2024c).	Variant-specific	F1; Precision; Recall
Logic, Puzzles, and Code			
ARC-AGI-1 (Chollet, 2019)	Grid-transformation puzzles requiring rule induction from a few input-output examples.	400 training / 400 public evaluation tasks	Task-level exact-match accuracy (two attempts per test input)
BBH (Suzgun et al., 2023)	A suite of 23 challenging BIG-Bench tasks covering diverse multi-step reasoning, language understanding, and knowledge.	23 tasks	Exact-match accuracy
T4D (Thinking for Doing) (Zhou et al., 2024b, 2023b)	Social reasoning requiring action selection based on inferred mental states.	Not reported as a stable benchmark-wide count	Action-selection accuracy
HumanEval / HumanEval+ (Chen et al., 2021; Liu et al., 2023b)	Python function-synthesis problems evaluated by unit tests; HumanEval+ retains the same problems with substantially expanded tests.	164 problems	pass@k
MBPP / MBPP+ (Austin et al., 2021; Liu et al., 2023b)	Short Python-program synthesis tasks; MBPP+ uses a curated subset of sanitized MBPP with substantially expanded tests.	MBPP: 974 total / 500 test; MBPP+: 378 tasks (EvalPlus v0.2.0)	pass@k
LiveCodeBench (Jain et al., 2024)	Continuously updated competitive-programming problems from dated releases; post-cutoff subsets help assess contamination risk.	Versioned rolling release	pass@1; pass@5
Knights & Knaves (Xie et al., 2025)	Procedurally generated truth-teller/liar puzzles with adjustable difficulty; train/test separation depends on generator settings.	Procedural	Accuracy
ZebraLogic (Lin et al., 2025)	Logic-grid puzzles with controllable complexity, designed to study scaling on multi-step constraint reasoning.	1,000 puzzles	Puzzle-level accuracy
Composition, Length, and Systematic Generalization			
SCAN (length split) (Lake and Baroni, 2018)	Synthetic command-to-action mapping; the length split tests generalization to longer output-action sequences.	16,990 train / 3,920 test	Exact Match
CFQ (Keysers et al., 2020)	Semantic parsing from English questions to SPARQL queries; MCD splits hold out novel compositions of familiar primitives.	239,357 total examples; each MCD split: 95,743 train / 11,968 validation / 11,968 test	Accuracy (exact-query correctness)
COGS (Kim and Linzen, 2020)	Sentence-to-logical-form mapping that tests novel combinations of familiar words and structures.	24,155 train / 3,000 dev / 3,000 ID test / 21,000 generalization	Exact Match
CommaQA-E (Chen et al., 2024)	Synthetic multi-hop question answering over fictional entities; evaluated for compositional generalization in Skills-in-Context.	Protocol-dependent	Exact Match (SKiC protocol)
Last-letter concatenation (Zhou et al., 2023a)	Concatenates the final letters of a word list; evaluation at increasing list lengths tests length generalization.	Protocol-dependent; the cited protocol tests lengths 4, 6, 8, 10, and 12	Exact Match

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Table 10 (continued)

Dataset / Benchmark	Characteristics and Evaluation Use	Dataset Size / Subset	Evaluation Metric
CLRS-30 (Veličković et al., 2022)	Thirty classical algorithms with structured inputs and algorithmic traces; tests generalization to larger problem sizes. Originally designed for neural algorithmic reasoning.	1,000 train trajectories per algorithm at size 16; evaluation counts use algorithm-specific multipliers	Micro-F1 (original benchmark evaluation)
MathGAP (Opedal et al., 2025)	Generated arithmetic word problems with controllable proof-tree structure and sentence ordering, enabling OOD evaluation.	Generated; test-set size depends on the experiment	Accuracy
Multi-digit arithmetic (Zhou et al., 2024d; Malach et al., 2026)	Addition and multiplication across digit lengths. Zhou et al. report addition generalization up to 100 digits in the main setup; Malach et al. study tool-augmented length generalization up to 1,000 digits.	Protocol-dependent; up to 100 digits (Zhou et al.) and 1,000 digits (Malach et al.)	Exact-Match Accuracy (Zhou et al.); Exact-Recovery Accuracy (Malach et al.)
Dynamic programming: maximum non-adjacent sum (Chen et al., 2024)	Sequence optimization that selects a maximum-sum subsequence with no adjacent elements; longer sequences test generalization beyond demonstrations.	Demonstrations: lengths 4–5; evaluation: lengths 4–8	Accuracy
Compositional-ARC (Mondorf et al., 2025)	ARC-derived episodes testing systematic generalization to held-out compositions of geometric transformations.	82,908 train / 8,546 validation / 8,546 test episodes (seed-1860 split)	Exact-Match Accuracy (per query grid)
OMEGA (Sun et al., 2025)	Generated math problems evaluating exploratory, compositional, and transformative generalization across mathematical domains.	Generated; protocol-dependent	Accuracy
XDomainBench (Gong et al., 2026)	Interactive, multi-turn benchmark for interdisciplinary scientific knowledge composition.	8,598 sessions across 20 scientific domains	Recall / F1 (turn-level); SessionSuccess@ τ (session-level)
Reversal Curse (Berglund et al., 2024)	Tests whether learning a relation in one direction ("A is B") supports reverse queries ("B is A"); the cited synthetic-fact setup is not the paper's only evaluation.	Protocol-dependent	Exact Match (synthetic-fact evaluation)
Tool Use, Planning, Agents, and Modality Shifts			
A-OKVQA (Schwenk et al., 2022)	Visual question answering requiring image content and outside knowledge; can probe modality shift when compared against a text-only reference.	24,903 examples in the original dataset; evaluated subsets depend on the protocol	VQA soft accuracy (direct-answer); accuracy (multiple-choice)
Tower of Hanoi	Planning puzzle with an optimal solution of $2^n - 1$ moves. MAKER supplies a strategy for reliable long-horizon execution; Malach et al. study length generalization with tool-augmented models.	Protocol-dependent	Zero-error completion (MAKER); protocol-specific metric for length generalization
GSM8K-XL (Alazraki and Rei, 2025)	GSM8K questions with magnified numbers, used to study tool use under magnitude shift.	Protocol-dependent	Exact-match accuracy of the final answer
FuncQA (Hao et al., 2023)	Numeric questions requiring chains of arithmetic function calls.	Protocol-dependent	Exact-match accuracy
MINTQA (Qian et al., 2025)	Multi-hop question answering used as an out-of-distribution evaluation of tool-use decisions; the benchmark targets new and long-tail knowledge.	Protocol-dependent	Answer accuracy; average tool use
SMART task families (Math, Time, Intention) (Qian et al., 2025)	Tool-use restraint across Math, Time, and Intention; the Intention domain is adapted from IN3.	Protocol-dependent	Accuracy and tool-use frequency; Intention also uses task-specific metrics
WebArena (Zhou et al., 2024c)	Interactive tasks on simulated websites. ReasoningBank evaluates cross-site memory transfer on a protocol-specific "Multi" subset.	812 tasks in the original benchmark; 684 tasks in the ReasoningBank protocol, including 29 Multi tasks	Task Success Rate

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Table 10 (continued)

Dataset / Benchmark	Characteristics and Evaluation Use	Dataset Size / Subset	Evaluation Metric
Mind2Web (Deng et al., 2023)	Web-navigation tasks with cross-task, cross-website, and cross-domain test settings.	Protocol-dependent	Step Success Rate; task-level Success Rate
SWE-bench Verified	Human-validated subset of real GitHub issues (Jimenez et al., 2024; OpenAI, 2024); task difficulty alone does not establish a distribution shift.	500 instances	Resolve rate
Surface Perturbation and Distractors			
GSM-IC (Shi et al., 2023)	GSM8K-style problems with irrelevant context that preserves the gold answer; tests distractibility rather than a train/test OOD split.	58,052 total examples; primary evaluation uses GSM-IC-4K	Micro Accuracy; Macro Accuracy; Normalized Accuracy
GSM-Plus (Li et al., 2024b)	Eight perturbations per GSM8K seed problem, covering numerical, arithmetic, understanding, distractor, and critical-thinking changes.	10,552 variants (1,319 seeds $\times$ 8); protocol may use subsets	Accuracy; Performance Drop Rate (PDR); Accurately Solved Pairs (ASP)
GSM-Symbolic (Mirzadeh et al., 2025)	Template-based numerical variants of grade-school math problems, enabling controlled evaluation across instantiations.	100 templates $\times$ 50 instances per variant set	Accuracy across instantiations (mean $\pm$ std.)
SVAMP (Patel et al., 2021)	Perturbed arithmetic word problems designed to test sensitivity to problem structure and surface cues; used as a zero-shot target in Gao et al. (2025b).	1,000 problems in the full dataset; evaluated subset is protocol-dependent	Accuracy
Recent and Contamination-Limited Benchmarks			
GSM1K (Zhang et al., 2024b)	Newly authored grade-school math problems modeled on GSM8K to study overfitting and potential contamination; public release limits future contamination claims.	1,205 problems	Accuracy
LiveBench (White et al., 2024)	Regularly refreshed questions designed to limit contamination; results should be interpreted relative to the model's likely training exposure and the benchmark release.	Periodically updated; report the release used	Task-specific objective scores; aggregate score
General365 (Liu et al., 2026a)	General-reasoning problems with generated variants that alter surface semantics or constraints while aiming to preserve core reasoning skills; variants share seeds.	365 seeds + 1,095 variants	Accuracy
EsoLang-Bench (Sharma and Chopra, 2026)	Programming problems in five esoteric languages; low public-data availability is a proxy for limited exposure, not proof of no exposure.	80 problems per language across five languages	Program accuracy via interpreter-based test-case verification

Table 10: Evaluation benchmarks referenced in this survey, grouped by evaluation setting. Counts and metrics follow the cited release or the studies that report them; "Protocol-dependent" means the value is set by individual studies. Whether an evaluation is in-distribution or out-of-distribution depends on a model's training exposure under a specific protocol, not on the benchmark name.

Method / paper	Evaluation evidence	Reported measurements	Generalization evidence and caveat
Trajectory restructuring			
Least-to-Most prompting (L2M) (Zhou et al., 2023a)	Last-letter/list concatenation, SCAN length splits, and GSM8K grouped by solution-step count.	Accuracy by list length or step count.	**Structural extrapolation.** Evaluates extrapolation relative to shorter in-context exemplars. Extrapolation across semantic domains is not assessed.
Divide-and-Conquer (DaC) prompting (Zhang et al., 2024c)	HaluEval-Summary, a constructed SciFact misinformation variant, and long-integer arithmetic tasks.	Accuracy, F1, precision, recall.	**Task performance.** Evaluates task performance without an explicit ID/OOD comparative framework.
MAKER (Meyerson et al., 2025)	Tower of Hanoi with a supplied strategy, including a 20-disk zero-error system run.	Zero-error completion; projected per-model error and cost estimates.	**Scaled-task performance.** Evaluates performance on scaled tasks; a formalized ID/OOD split is not established.
SKiC (Chen et al., 2024)	Synthetic structural tasks (last-letter concatenation, arithmetic digit-length extrapolation, dynamic programming) and standard reasoning evaluations (CommaQA-E, MATH).	Accuracy/Exact Match by task size and prompt condition; internal-skill activation rates.	**Structural extrapolation.** Demonstrates length/depth extrapolation in synthetic settings. Standard reasoning benchmark evaluations lack a paired ID/OOD split.
Constraint-First Reasoning (CFR) (Ma et al., 2026)	AIME24/25/26, CMIMC25, BRUMO25, and OlympiadBench.	Pass@1 averaged over four runs (`avg@4`); accuracy and token overhead.	**Task performance and format robustness.** Reports quantitative gains on competition math tasks without a predefined ID/OOD baseline.
Self-Discover (Zhou et al., 2024b)	BBH, T4D, and MATH, including transfer across model families.	Task-level accuracy gains.	**Task performance and cross-model transfer.** Evaluates task performance and transfer across model families; a data-distribution OOD test is not established.
Self-Demos (He et al., 2024b)	Query-aware demonstrations and the OOD-Tool set.	Task accuracy compared with few-shot baselines.	**Out-of-demonstration evaluation.** Evaluates capabilities beyond the provided demonstrations; shifts from the model's pre-training distribution are not explicitly tested.
Compute scaling			
ReST-MCTS* (Zhang et al., 2024a)	MATH-sourced training and evaluation settings compared with non-source benchmarks (GPQA-Diamond, CEval-Hard).	Accuracy and average scores for PRM-guided MCTS and self-training.	**Cross-benchmark evaluation.** Reports comparative gains on non-source benchmarks. Base-model pre-training exposure is unquantified.
Boosting PRMs with MCTS (Chan et al., 2025)	Math, code, and knowledge benchmarks evaluating MCTS-generated training data and inference scaling (Best-of-N, beam search).	Benchmark scores and scaling trends under sample/search budgets.	**Scaling behavior.** Evaluates scaling behavior and improvements from generated training data without a controlled ID/OOD test.
Compute-optimal test-time scaling (Snell et al., 2024)	MATH test set stratified into difficulty bins to evaluate search and sequential revision strategies.	MATH accuracy versus generation/compute budget.	**Difficulty scaling.** Stratifies evaluation by task difficulty; a formal ID/OOD distribution split is not constructed.
Test-Time Scaling via Error Localization (TTEL) (Chitale et al., 2026)	MATH-500, AIME24/25, GPQA-Diamond, and LiveCodeBench evaluating localized regeneration.	Pass@k and token budgets under rollout/sample constraints.	**Localized revision.** Evaluates pass@k improvements under localized revision; explicit train/test distribution shifts are not assessed.

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Table 11 (continued)

Method / paper	Evaluation evidence	Reported measurements	Generalization evidence and caveat
State assessment			
Residual-stream trajectory detector (Damirchi et al., 2026)	ARC-C (reference) and other reasoning/commonsense datasets (targets) used for cross-benchmark candidate detection.	AUROC and correct-candidate detection rates.	**Assessment transfer.** Evaluates transferability of the assessment/detection signal to held-out benchmarks.
UPAIR (Yan et al., 2026)	Diagnostic experiments and selective interventions evaluated on MATH-500, AIME24/25, GPQA-Diamond, and LiveCodeBench.	Final accuracy/token cost; partial diagnosis and intervention metrics.	**Diagnostic and intervention performance.** Evaluates diagnostic signals and intervention benefits without a controlled OOD comparison.
Thought Calibration (Wu et al., 2025)	Hidden-state probes and risk-controlled halting evaluated on target sets (AIME24, GPQA, MATH-500), using s1K-1.1 as reference data.	Probe AUROC, stopping/risk statistics, task accuracy, and step/token savings.	**Calibration and task performance.** Assesses calibration and task gains on target sets; calibration stability under distribution shift is measured separately.
Online Reasoning Calibration (ORCA) (Zhou et al., 2026)	Test-time online calibration evaluated on MATH-500, AIME24/25, GPQA-Diamond, and LiveCodeBench.	Accuracy, generated tokens, and risk/calibration statistics.	**Online calibration.** Explicitly evaluates calibration under target settings. The extent of distribution shift depends on the evaluation protocol.
Generative verifier / RL^V (Sareen et al., 2025)	Candidate selection evaluated on MATH-500, harder MATH^2, and a physics set following MATH training.	Candidate-selection accuracy under parallel test-time scaling.	**Task generalization.** Demonstrates downstream selection capabilities on math-adjacent and physics tasks.
Verifier-Driven Sample Selection TTT (VDS-TTT) (Moradi et al., 2025)	High-confidence outputs used as pseudo-labels for generator adaptation on GSM8K, MATH-500, and AIME.	Exact Match/accuracy against a frozen base and verifier-only baselines.	**Target-task adaptation.** Evaluates target-task gains from generator adaptation; a paired reference distribution is not defined.
Retrieval-Augmented PRM (RetrievalPRM) (Zhu et al., 2025a)	ProcessBench (GSM8K, MATH, OlympiadBench, Omni-MATH) evaluated using a PRM800K and Math-Shepherd retrieval pool.	ProcessBench F1 and ArithACC for first-error detection.	**Question-level generalization.** Competition subsets are evaluated as question-OOD relative to PRM training. The scope of this claim depends on the training and evaluation split.
Inference-time externalization			
Interactive tool-use memory for bounded-state models (Malach et al., 2026)	Interactive memory tools evaluated on synthetic addition/multiplication (up to 1,000 digits), logical graphs, and Tower of Hanoi.	Exact-recovery and task accuracy by sequence size.	**Structural length generalization.** Evaluates length generalization for specific architectures; semantic or algorithmic OOD is not tested.
Chain-of-Abstraction (CoA) (Gao et al., 2025b)	Zero-shot evaluations on math datasets (GSM8K, SVAMP, MAWPS) and knowledge-intensive QA.	Exact Match/accuracy, arithmetic error rate, and latency.	**Cross-task transfer.** Evaluates transfer to designated target datasets. Dataset overlap with pre-training is not assessed.
TECTON (Alazraki and Rei, 2025)	Tool selection evaluated on GSM8K-XL and designated math sets with magnitude/dataset variation.	Final task accuracy.	**Tool-selection transfer.** Demonstrates transfer across math word problems featuring magnitude shifts.
SMART / SMARTAgent (Qian et al., 2025)	Tool-use decisions evaluated across task types, including GSM8K, MINTQA, and Math/Time-style tasks.	Task accuracy, tool calls, tool-overuse rate, and decision confidence.	**Cross-task tool-use evaluation.** Evaluates tool-use restraint and transfer across task types.
ReasoningBank (Ouyang et al., 2026)	Task-stream memory accumulation evaluated on Mind2Web, WebArena Multi, and SWE-Bench-style tasks.	Success rate, average interaction steps, and token cost.	**Within-stream improvement.** Demonstrates strategy-level memory and within-stream improvements; an isolated held-out target is not established.

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Table 11 (continued)

Method / paper	Evaluation evidence	Reported measurements	Generalization evidence and caveat
MILES (Tong and Gong, 2026)	Modular sub-instruction selection evaluated on MATH-500, AIME24/25, GPQA-Diamond, and LiveCodeBench.	Majority-vote accuracy, sub-instruction rates, memory size, and accuracy–cost frontier.	**Accuracy–cost trade-offs.** Evaluates accuracy–cost trade-offs; cross-domain memory transfer is not directly assessed.

Table 11: Evaluation evidence for inference-time methods. The table summarizes evaluation protocols, reported results, and the strengths and limitations of generalization evidence. Family headings group methods by their primary intervention. Benchmark transfer, difficulty scaling, or performance gains alone do not establish OOD generalization; conclusions depend on evaluation protocols and known training exposure.

Metric	Definition	Interpretation	Comparability considerations
Task performance			
Accuracy / Exact Match	For N problems and a task-specific correctness checker V_q, $\mathrm{Acc} = \frac{1}{N}\sum_{q=1}^{N} V_q(\hat{y}_q).$ Exact Match applies the benchmark's answer-matching rule.	Measures the proportion of correct outputs; higher is better.	Requires compatible answer parsing, normalization, and test subsets.
Success Rate	For N tasks, $\mathrm{SR} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[\text{task } i \text{ succeeds}]$.	Measures the proportion of tasks completed successfully.	The benchmark's task-level success criterion must be consistent.
Sampling and selection protocols			
Average accuracy (avg@k)	Given k sampled responses per problem, $\mathrm{avg@}k = \frac{1}{N}\sum_{q=1}^{N}\frac{1}{k}\sum_{s=1}^{k} V_q(\hat{y}_{q,s}).$	Measures mean single-sample accuracy across k responses; does not measure whether at least one response is correct.	Use the same k, decoding settings, correctness checker, and evaluation set for comparisons.
pass@k	Given $n \geq k$ sampled candidates per problem, of which c_q are correct, $\widehat{\mathrm{pass@}k} = \frac{1}{N}\sum_{q=1}^{N}\left[1 - \frac{\binom{n-c_q}{k}}{\binom{n}{k}}\right],$ where $\binom{n-c_q}{k} = 0$ when $n - c_q < k$.	Estimates the fraction of problems for which at least one of k candidates drawn without replacement from the n_i generated candidates is correct; non-decreasing in k.	Match k, candidate counts, decoding settings, and compute budget.
Majority voting (maj@k)	Selects the most frequent normalized answer among k sampled candidates and evaluates the selected answer (Wang et al., 2023).	Measures the effectiveness of answer aggregation across samples.	Depends on sample count, answer normalization, and tie-breaking rules.
Best-of-N selection	Generates N candidates, selects one using a scorer or verifier, and evaluates the selected output with a task metric.	Measures task performance under a candidate-selection protocol.	A selection protocol rather than a standalone metric. Specify N and the selection procedure.
Verification and discrimination			
F1	$\mathrm{F1} = 2\frac{\mathrm{Precision}\cdot\mathrm{Recall}}{\mathrm{Precision}+\mathrm{Recall}}$, defined as 0 when $\mathrm{Precision} + \mathrm{Recall} = 0$.	Summarizes precision and recall; higher is generally better.	Depends on the positive-class definition, averaging method, and zero-denominator convention.

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Table 12 (continued)

Metric	Definition	Interpretation	Comparability considerations
AUROC	Area under the receiver operating characteristic curve; measures how well a score ranks positive instances above negative ones.	Higher values indicate better discrimination.	Does not measure probability calibration; specify the positive class.
Efficiency and resource cost			
Tokens, latency, and compute budget	Operational costs such as generated tokens, wall-clock time, model calls, and allocated computation.	Characterizes the resources required to obtain a result.	Compare costs alongside task performance under compatible hardware, settings, and budgets.
Survey-defined quantities for generalization analysis			
Generalization gap Δ	$\Delta = \mathrm{Acc}_{\mathrm{ref}} - \mathrm{Acc}_{\mathrm{shift}}$, computed on matched reference and shifted evaluation sets.	A positive value indicates a performance drop under the shift; a negative value indicates an improvement.	Requires a comparable model, checkpoint, metric, and evaluation protocol. Interpret alongside both scores.
Relative drop Δ_{rel}	$\Delta_{\mathrm{rel}} = \frac{\Delta}{\mathrm{Acc}_{\mathrm{ref}}}$, defined when $\mathrm{Acc}_{\mathrm{ref}} > 0$.	Expresses the signed performance gap relative to reference accuracy.	Report alongside the absolute gap and both underlying scores.
Gain over baseline G	$G = \mathrm{Score}_{\mathrm{method,target}} - \mathrm{Score}_{\mathrm{baseline,target}}$ under the same target protocol.	For metrics where higher is better, positive values indicate an improvement over the baseline on that target.	Not equivalent to a generalization gap; by itself, it does not establish generalization beyond training exposure.

Table 12: Metrics and protocols for evaluating reasoning generalization. Cross-study comparisons require aligned evaluation protocols and resource budgets; ID/OOD status depends on evaluation conditions and training exposure, not benchmark names alone.